

Exercises: Simple Gaussian Linear Regression Part II

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(Thanks to Dr. Roberto Ascari for providing these exercises)

(Referring to the theoretical parts: 4, 5, 6, 7)

2 Computer repair data

A computer repair company is interested in knowing the relationship between the duration of interventions (measured in minutes) and the number of electronic components to be replaced or repaired. Therefore, a simple linear regression model was considered to explain the duration in minutes of interventions (y) as a function of the number of units (x) to be replaced.

A sample of 14 interventions provided the following data: $\bar{y} = 95.768$, $\bar{x} = 6$, $\sum_{i=1}^{14} (y_i - \bar{y})^2 = 31108.357$, and $\sum_{i=1}^{14} (x_i - \bar{x})^2 = 114$. The model provides a coefficient of determination $R^2 = 0.984$.

Exercise 2.1

Starting from the data, compute the maximum likelihood estimates of β_1 and β_2 . Then, write the equation of the estimated linear regression model.

• SIMPLE LINEAR REGRESSION MODEL

$$Y_i = \beta_1 + \beta_2 x_i + \varepsilon_i$$

↑ TIME DURATION ↑ UNITS TO BE REPAIRED

ASSUMPTIONS $i = 1, \dots, n$

- $E(\varepsilon_i) = 0 \longrightarrow E(Y_i) = \beta_1 + \beta_2 x_i$
- $\text{VAR}(\varepsilon_i) = \sigma^2 > 0$
- $\text{COV}(\varepsilon_i, \varepsilon_k) = 0$ FOR $i \neq k \longrightarrow \text{COV}(Y_i, Y_k) = 0$
- x_i : FIXED AND NON STOCHASTIC
- x_i : CAN NOT BE ALL EQUAL

- $n = 14$
- $\bar{y} = 95.768$
- $\bar{x} = 6$
- $\sum (y_i - \bar{y})^2 = 31108.357$
- $\sum (x_i - \bar{x})^2 = 114$

$$R^2 = 0.984$$

$$\hat{\beta}_2 = \frac{S_{xy}}{S_{xx}} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{\sum x_i y_i - n \bar{x} \bar{y}}{\sum x_i^2 - n \bar{x}^2} = \frac{143.69}{114} = 1.2604386$$

$$\hat{\beta}_1 = \bar{y} - \hat{\beta}_2 \bar{x}$$

FOR SIMPLE MODEL ONLY

$$R^2 = r_{xy}^2 \quad \text{AND} \quad R^2 = \left(\frac{\beta_2}{\sqrt{\frac{S_{xx}}{S_{yy}}}} \right)^2$$

$$\hookrightarrow S_{xy} = \sqrt{R^2} S_{xx} S_{yy} = \sqrt{0.984} \cdot \sqrt{\frac{1}{n-1} 114} \cdot \sqrt{\frac{1}{n-1} 31108.357} = 143.69 \longrightarrow \beta_2 = \frac{S_{xy}}{S_{xx}} = \frac{143.69}{\frac{114}{13}} = 16.386$$

$n = 14 \quad 1$

$$\hat{\beta}_1 = \bar{y} - \hat{\beta}_2 \bar{x} = 95.768 - 16.38 \cdot 6 = -2.55 \longrightarrow Y_i = -2.55 + 16.38 X_i$$

Exercise 2.2

Find the estimate for the variance σ^2 using the decomposition of the total sum of squares.

Through a valid test, verify the goodness of fit at 5% confidence level.

$$s^2 = \frac{1}{n} SSE = \frac{1}{14} \left[\sum (y_i - \hat{y}_i)^2 - \beta_1 \sum (x_i - \bar{x})^2 \right] = \frac{1}{14} \left(31108.357 - (16.386)^2 \cdot 114 \right) = 35.66$$

$$s^2 = \frac{SSE}{n-p}$$

$$\bullet SST = 31108.357 \quad \rightarrow SSR = R^2 SST = 30670$$

$$\bullet n-p = 14-2 = 12$$

$$\bullet R^2 = 0.984 = \frac{SSR}{SST} \rightarrow R^2 = \frac{SST - SSE}{SST} \rightarrow -R^2 SST + SST = SSE \rightarrow SSE = -0.984 \cdot 31108.357 + 31108.357 = 497$$

$$s^2 = \frac{497.73}{12} = 41.47$$

GOODNESS OF FIT

$$\begin{cases} H_0: \beta_2 = 0 \\ H_1: \beta_2 \neq 0 \end{cases}$$

$$t_{obs}^{H_0} \sim t_{n-p} \quad t_{obs} = \frac{\hat{\beta}_2 - \beta_2}{\sqrt{VAR(\hat{\beta}_2)}} = \frac{16.386}{\sqrt{0.36}} = 27.31$$

$$VAR(\hat{\beta}_2) = \frac{s^2}{\sum (x_i - \bar{x})^2} = \frac{41.47}{114} = 0.36$$

$$\begin{cases} H_0: \beta_2 = 0 \\ H_1: \beta_2 \neq 0 \end{cases} \quad f_{obs}^{H_0} \sim f_{p-p_0, n-p} \quad \begin{matrix} 1 \\ \downarrow \\ p-p_0 \end{matrix} \quad \begin{matrix} 12 \\ \downarrow \\ n-p \end{matrix}$$

$$f_{obs} = \frac{\frac{SSR}{p-p_0}}{\frac{SSE}{n-p}} = \frac{\frac{30670}{1}}{\frac{497}{14-2}} = 739$$

$$\alpha^{obs} = P(f_{p-p_0, n-p} > f_{obs}) \approx 1 - \alpha \approx 0$$

SINCE $\alpha^{obs} < \alpha$ FOR $\alpha = 0.05$ I REJECT H_0
AT THIS CONFIDENCE LEVEL I PREFER
THE MODEL WITH H OF UNITS TO BE
REPLACED WRT TO THE MODEL W/O UT

$$\alpha^{obs} = P(|t_{n-p}| \geq |t_{obs}|) = 2 \min \{ P(t_{n-p} < -t_{obs}), P(t_{n-p} > t_{obs}) \}$$

$$\approx 2(1 - \alpha) \approx 0$$

I REJECT H_0 SINCE $\alpha^{obs} < \alpha$ I PREFER THE
MODEL WITH NUMBER OF UNITS TO BE
REPLACED WRT THE MODEL W/O UT

Exercise 2.3

Given the standard errors (S.E.) of the estimators $\hat{\beta}_1$ and $\hat{\beta}_2$, which correspond to $\sqrt{\widehat{Var}(\hat{\beta}_1)} = 4.014$ and $\sqrt{\widehat{Var}(\hat{\beta}_2)} = 0.604$. Through a valid test (at 5 % confidence level), verify if the coefficients β_1 and β_2 are significant (you can use the following t-table for computing p-values).

t Table

cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
one-tail	0.50	0.25	0.20	0.15	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
two-tails	1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.01	0.002	0.001
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073

$$\begin{cases} H_0: \beta_2 = 0 \\ H_1: \beta_2 \neq 0 \end{cases} \quad t_{obs}^{(2)} \sim t_{n-p} \quad t_{obs}^{(2)} = \frac{\hat{\beta}_2 - \beta_2}{\sqrt{\widehat{Var}(\hat{\beta}_2)}} = \frac{7.326}{\sqrt{0.96}} = 29.31$$

$$\alpha^{obs} = P(|t_{n-p}| \geq |t_{obs}|) = 2 \min \{ P(t_{n-p} < -t_{obs}), P(t_{n-p} > t_{obs}) \}$$

$$\approx 2(1 - \sim) \approx 0$$

I REJECT H_0 SINCE $\alpha^{obs} < \alpha$ I PREFER THE
MODEL WITH NUMBER OF UNITS TO BE
REDUCED WRT THE MODEL W/O UT

$$\begin{cases} H_0: \beta_1 = 0 \\ H_1: \beta_1 \neq 0 \end{cases} \quad t_{obs}^{(1)} \sim t_{n-p} \quad n-p = 2-12$$

$$t_{obs}^{(1)} = \frac{\hat{\beta}_1 - \beta_1}{\sqrt{\widehat{Var}(\hat{\beta}_1)}} = \frac{-2.55 - 0}{4} = -0.636$$

$$\alpha^{obs} = P(|t_{n-p}| \geq |t_{obs}|) = 2 \min \{ P(t_{n-p} < -t_{obs}), P(t_{n-p} > t_{obs}) \}$$

$$\approx 2(1 - 0.70) > 0.10$$

DO NOT REJECT H_0

EXTRA: COMPUTE $\widehat{Var}(\hat{\beta}_1)$

$$\widehat{Var}(\hat{\beta}_1) = s^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{\sum (x_i - \bar{x})^2} \right) = 41.47 \left(\frac{1}{14} + \frac{6^2}{414} \right) = 16.05 \rightarrow SE(\hat{\beta}_1) = 4.01$$

Exercise 2.4

Given the ex. 2.2, is there any statistical test in the exercise 2.3 that might be unnecessary?

Test on β_2

3 Bacteria mortality data

Suppose we want to analyze bacterial mortality (y) as a function of radiation exposure (x). The output of a linear regression of y as a function of x is partially summarized in the table below:

Table 1: Output of a linear regression.

Variable	Coefficients	S.E.	T-value	P-value
Constant	49.162	22.76	①	②
Exposure (x)	-19.46	③	-7.79	<0.0001

where $n = 15$, $R^2 = 0.823$ and $\hat{\sigma} = 41.83$.

Exercise 3.1

Complete the Table 1 writing the equations you should use.

$\downarrow$ MORTALITY $\downarrow$ EXPOSURE
 $Y_i = \beta_1 + \beta_2 x_i + \epsilon_i$

① $\begin{cases} H_0: \beta_1 = 0 \\ H_1: \beta_1 \neq 0 \end{cases} \quad \epsilon_1^{OBS} \sim \epsilon_{n-p} \quad \downarrow \text{75-2=73}$

$$\epsilon_1^{OBS} = \frac{\hat{\beta}_1 - \beta_1}{SE(\hat{\beta}_1)} = \frac{49.162 - 0}{22.76} = 2.160$$

② $\alpha^{OBS} = P(|\epsilon_{n-p}| > |\epsilon_1^{OBS}|) = 2 \min\{P(\epsilon_{n-p} < -\epsilon_1^{OBS}), P(\epsilon_{n-p} > \epsilon_1^{OBS})\}$
 $= 2(1 - 0.98) = 0.04$

③ $\begin{cases} H_0: \beta_2 = 0 \\ H_1: \beta_2 \neq 0 \end{cases} \quad \epsilon_{0.05}^2 \sim \epsilon_{n-p} \quad \downarrow \text{73} \quad \epsilon_2^{OBS} = -7.79 = \frac{\hat{\beta}_2 - \beta_1}{SE(\hat{\beta}_2)} \rightarrow SE(\hat{\beta}_2) = \frac{-19.46}{-7.79} = 2.493$

SIMPLE LINEAR REGRESSION

$E(\epsilon_i) = 0 \quad \forall i \rightarrow E(Y_i) = \beta_1 + \beta_2 x_i$

$VAR(\epsilon_i) = \sigma^2 > 0$

$COV(\epsilon_i, \epsilon_k) = 0$ for $i \neq k \rightarrow COV(Y_i, Y_k) = 0$

x_i : FIXED AND NON STOCHASTIC

x_i : CAN NOT BE ALL EQUAL

Exercise 3.2

Through a valid statistical test, evaluate the goodness of fit of the model.

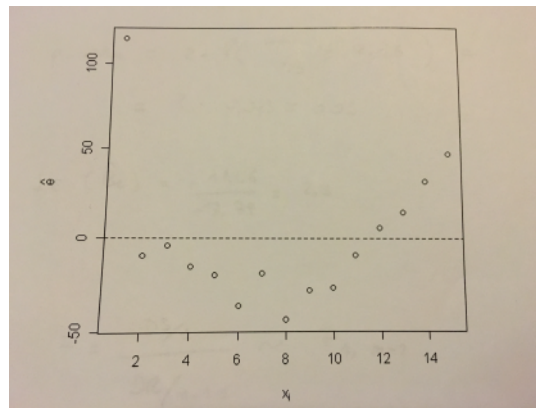
$$\begin{cases} H_0: \beta_3 = 0 \\ H_1: \beta_3 \neq 0 \end{cases} \quad E_2^{(H_0)} \sim N(0, \sigma^2)$$

CAN BE SKIPPED $\chi^2_2 < 0.0001$

SO REJECT $H_0 \rightarrow$ MODEL IS GOOD

Exercise 3.3

Discuss about the hypothesis related to the model, looking the following residual plot.



- TOO FEW N TO MAKE GOOD ASSUMPTIONS
- CHECK IF RESIDUES FOLLOW A NORMAL DISTRIBUTION
- SEEMS THERE IS A PATTERN IN RESIDUAL
 - TRANSFORM COVARIATES
 - ADD NEW COVARIATES
- ERROR SHOULD BE CLOUD DISPERSION AROUND 0

4 Grades data

In 2011 among 62 adolescents, the variables x "daily hours spent on average to video games" and y "average report card grade" were observed. We proposed a gaussian simple linear model with y , as response variable, and obtained the following estimates: $\hat{\beta}_1 = 7.4$, $\hat{\beta}_2 = -0.48$, $SSE = 223$, $\frac{Var(\hat{\beta}_1)}{\sigma^2} = 3.43$ and $\frac{Var(\hat{\beta}_2)}{\sigma^2} = 0.07$.

Exercise 4.1

Compute the OLS estimates for β_1 and β_2 and provide an explanation.

n=62

CARD GRADE $\rightarrow y$ *DAILY HOURS VIDEOGAMES* $\rightarrow x$

- $Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$
- $\hat{\beta}_1 = 7.4$
- $\hat{\beta}_2 = -0.48$
- $SSE = 223$
- $\frac{Var(\hat{\beta}_1)}{\sigma^2} = 3.43 \rightarrow SE(\hat{\beta}_1) = \sqrt{3.43 \cdot s^2} = \sqrt{3.43 \cdot 3.71} = 3.57$
- $\frac{Var(\hat{\beta}_2)}{\sigma^2} = 0.07 \rightarrow SE(\hat{\beta}_2) = \sqrt{0.07 \cdot s^2} = \sqrt{0.07 \cdot 3.71} = 0.509$
- $s^2 = \frac{SSE}{n-p} = \frac{223}{60} = 3.71$

ASSUMPTIONS

- $E(\varepsilon_i) = 0 \quad \forall i = 1, \dots, n$
- $Var(\varepsilon_i) = \sigma^2 > 0$
- $Cov(\varepsilon_i; \varepsilon_k) = 0$ for $i \neq k \rightarrow Cov(Y_i; Y_k) = 0$
- x_i : FIXED AND NON-STOCHASTIC
- x_i : CAN NOT BE ALL EQUAL

Exercise 4.2

Through a valid test, use p-values to evaluate if the coefficients are significant (you can use the t-table below for computing p-values). Then, evaluate the goodness of fit using p-values.

26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.000	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390

$$\begin{cases} H_0: \beta_1 = 0 \\ H_1: \beta_1 \neq 0 \end{cases} \quad \epsilon_1^{OBS} \sim \epsilon_{n,p} \quad \epsilon_1^{OBS} = \frac{\hat{\beta}_1 - \beta_1}{SE(\hat{\beta}_1)} = \frac{7.4 - 0}{3.57} = 2.07$$

$$\alpha^{OBS} = P(\epsilon_{n,p} | \epsilon_1^{OBS}) = 2 \min \{ P(\epsilon_{n,p} < -\epsilon_1^{OBS}); P(\epsilon_{n,p} > \epsilon_1^{OBS}) \} \sim 2(1 - \alpha) \sim 0$$

I REFUSE TO H_0

$$\begin{cases} H_0: \beta_2 = 0 \\ H_1: \beta_2 \neq 0 \end{cases} \quad \epsilon_2^{OBS} \sim \epsilon_{n,p} \quad \epsilon_2^{OBS} = \frac{\hat{\beta}_2 - \beta_2}{SE(\hat{\beta}_2)} = \frac{-0.18 - 0}{0.509} = -0.94$$

$$\alpha^{OBS} = P(\epsilon_{n,p} | \epsilon_2^{OBS}) = 2 \min \{ P(\epsilon_{n,p} < -\epsilon_2^{OBS}); P(\epsilon_{n,p} > \epsilon_2^{OBS}) \} \sim 2(1 - \alpha) \sim 0.35$$

Since $\alpha^{OBS} > \alpha$ for $\alpha = 0.1$; $\alpha = 0.05$; $\alpha = 0.01$ I CAN NOT REFUSE $H_0 \rightarrow$ Model NOT GOOD

5 Additional exercise

A linear regression model was estimated on 82 units. Complete the tables below and specify the hypothesis, the test statistic and p-value for inference.

Table 2: Analysis of variance.

	Deviance	Sum of squares	d.o.f.	F	p-value
SSR	Residual	3589.6	1	10.21	
SSE	Regression		-	-	-
SST	Total		-	-	-

$$S_{OBS} = \frac{SSR}{p-1} \rightarrow SSE = \frac{SSR}{p-1} = \frac{3589.6}{82-2} = 28.126$$

$$SST = SSE + SSR = 28126 + 3589.6 = 31715$$

where d.o.f. means degree of freedom.

Table 3: Output of a linear regression.

Variable	Coefficients	S.E.	T-value	P-value
Constant	12.7		0.82	
x_1	-19.3			0.002

$$Y_i = \beta_1 + \beta_2 x_i$$

① $\begin{cases} H_0: \beta_1 = 0 \\ H_1: \beta_1 \neq 0 \end{cases}$ $\epsilon_1^{OBS} \sim \epsilon_{n-p}$

$$\epsilon_1^{OBS} = \hat{\beta}_1 - \beta_1 \rightarrow SE(\hat{\beta}_1) = \frac{12.7}{0.82} = 15.49$$

② $\alpha^{OBS} = P(|\epsilon_{n-p}| > |\epsilon_1^{OBS}|) = 2 \cdot P(\epsilon_{n-p} < -\epsilon_1^{OBS}) = 2(1 - 0.8) \hat{=} 0.4$

③ $\begin{cases} H_0: \beta_2 = 0 \\ H_1: \beta_2 \neq 0 \end{cases}$ $\epsilon_{n-p}^{OBS} \sim \epsilon_{n-p}$

$$\alpha^{OBS} = 0.002 = 2(1 - \dots) \rightarrow \epsilon_{n-p} = -3.195$$

④ $SE(\hat{\beta}_2) = \frac{\hat{\beta}_2}{\epsilon_{n-p}^{OBS}} = \frac{-19.3}{-3.195} = 6.04$

$$E(\epsilon_i) = 0 \quad \forall i = 1, \dots, n \rightarrow E(Y_i) = \beta_1 + \beta_2 x_i$$

$$VAR(\epsilon_i) = \sigma^2 > 0 \quad \forall i = 1, \dots, n$$

$$Cov(\epsilon_i, \epsilon_k) = 0 \quad \text{for } i \neq k \rightarrow Cov(Y_i, Y_k) = 0$$

$$x_i: \text{FIXED AND NON-STOCHASTIC}$$

$$x_i: \text{CAN NOT BE RE-ENTERED}$$